

Viraj Thakkar

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RESEARCH INTERESTS

Data Infrastructure: Key-Value stores (RocksDB); Timeseries databases (InfluxDB); Efficient Storage for Quantum Computing
Smart Storage: AI/ML for storage system; Real-Time Storage Systems for high-velocity and high-volume data

EDUCATION

Ph.D. Computer Science	Aug 2023 - Ongoing
Arizona State University, Tempe, AZ	3.92 / 4.00
M.S. Computer Science	Jan 2022 – Aug 2023
Arizona State University, Tempe, AZ	3.87 / 4.00
B. Tech. Information Technology	July 2017 – Aug 2021
K. J. Somaiya College of Engineering, Mumbai, India	8.56 / 10.0

PROJECTS

LLM-Assisted Tuning for LSM-KVS C++, RocksDB, Docker, Python	12/2023 - Ongoing
- Lead a team of undergraduate and graduate students to design and develop a framework for automated prompt generation and feedback loop for generating configuration files for. - Designed and Implemented architecture for multiple consistency levels within same database instance at the same time. - Present the project at HotStorage 2024 and won the Best Paper award at the conference.	
Smart Consistency in Distributed KV-Stores C++, HDFS, RocksDB, Kubernetes, Docker, ML	02/2022 – 08/2023
- Developed a scalable, distributed and storage-disaggregated Key-Value database service using RocksDB, HDFS, Docker, and Kubernetes. - Designed and Implemented architecture for multiple consistency levels within same database instance at the same time. - Currently implementing an ML model to automate assignment of consistency level to KV-pairs.	
Memories: Image Segmentation Library CI/CD, Python, OpenCV, PIL	08/2021 – 03/2022
- Creator and maintainer for the 'Memories' python library for Image Segmentation and Metadata management - Achieved over 14,000 downloads till date.	
Project Chakravyuh DevOps (Azure, Docker), scikit-learn, pandas, OpenCV, Docker	09/2018 – 10/2020
- Led and developed a team of six who developed a compartmentalized system for dark-web crawling with capabilities of text-analysis, media-analysis, form-filling, website monitoring. - Developed media-analysis module that tags any visual media based on its content using metadata and computer vision. - Designed and implemented an automated form-filling module that facilitates access of content hidden behind login/sign-up prompts without user-interaction.	

EXPERIENCE

Arizona State University: Research Assistant	08/2023 - Ongoing
- Working on 3 major projects under Dr. Zhichao Cao towards completion of my PhD. - Learning basics of Quantum Computing and possible research opportunities from domains of Storage and Memory.	
Amazon: Software Development Engineer Intern	05/2023 – 08/2023
- Set up product documentation to seamlessly onboard future team members. - Worked with the team to develop a full-stack application (using AWS Lambda, DynamoDB, S3 buckets, ECS, SQS CloudFormation) that provides an interface to check for copyright infringements for products on Amazon. - Completed internal courses for AutoGluon and Foundational ML courses.	
Arizona State University: Graduate Service Assistant	08/2022 – 05/2023
- Grader for CSE-330 and CSE-511 for Fall 2022 and Spring 2023 semesters under Prof. Zhichao Cao. - Implemented and assisted in re-designing course assignments and projects for CSE-511	
Cyber Peace Foundation, Mumbai, IN: Analyst (Intern)	03/2020 - 08/2020
- Developed automated crawling and investigating pipeline in 2 months for abusive content on various social media. - Team Leader for creation of detection and reporting tools for fake/malicious COVID websites. The tool successfully analyzed over 10,000 websites, identifying, and reporting 503 websites as malicious.	

KEM Teaching Hospital, Mumbai, IN: Developer (Intern)

05/2020 - 08/2020

- Implemented and deployed an interactive dashboard for KEM Hospital to monitor patients in the COVID-19 pandemic.
- Tool used for 7+ months in the pandemic, assisting in management and cleaning 5000+ patient records to create a centralized hospital records database.

Somaiya Vidyavihar, IN & CIFF (Children's Investment Fund Foundation), UK: Junior Researcher

09/2019 - 03/2020

- Worked on how technology has brought about Online Child Sexual Abuse and Exploitation (OCSA & OCSE).
- Developed tool to automate search and filter instances of OCSE and OCSE in Legal Cases

PUBLICATIONS

[HotStorage '24] Viraj Thakkar, Madhumitha Sukumar, Jiaxin Dai, Kaushiki Singh, and Zhichao Cao. 2024. "Can Modern LLMs Tune and Configure LSM-based Key-Value Stores?" *In Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems, July 08, 2024. ACM, Santa Clara CA USA, 116–123. [URL](#). BEST PAPER AWARD.*

[SIGMOD '24] Qiaolin Yu, Chang Guo, Jay Zhuang, **Viraj Thakkar**, Jianguo Wang, Zhichao Cao. "CaaS-LSM: Compaction-as-a-Service for LSM-based Key-Value Stores in Storage Disaggregated Infrastructure". *Proceedings of ACM Conference on Management of Data (SIGMOD), 2024, [URL](#).*

[Book: RPA '23] Ashwini Dalvi, **Viraj Thakkar**, Smit Moradiya, Aditya Vedpathak, Irfan Siddavatam, Faruk Kazi, S. G. Bhirud. "FLASH: Web-Form's Logical Analysis & Session Handling Automatic Form Classification and Filling on Surface and Dark Web". *Robotic Process Automation (2023): 61-100. [URL](#)*

[Book: HRAI-HCI '22] Irfan Siddavatam, Ashwini Dalvi, Abhishek Patel, Aditya Panchal, Aditya S. Vedpathak, **Viraj Thakkar**. "ML-Enabled Informed Intervention for Crowdsourcing-Based Optimization of Medical Resources." *Handbook of Research on Applied Intelligence for Health and Clinical Informatics*, pp. 291-315. IGI Global, 2022. [URL](#)

[ICACIT '21] Ashwini Dalvi, Irfan Siddavatam, **Viraj Thakkar**, Aditya Vedpathak, Abhishek Patel. "Recognizing Child Unsafe Apps Through User Reviews on the Google Play Store." *Advanced Computing and Intelligent Technologies* (pp. 111-120). Springer, Singapore 2022. [URL](#)

[ICAST '21] Irfan Siddavatam, Ashwini Dalvi, **Viraj Thakkar**, Aditya Vedpathak, Smit Moradiya, Apoorva Jain. "Explainability using decision trees and Monte Carlo simulations." *Proceedings of the 4th International Conference on Advances in Science & Technology (ICAST2021). [URL](#)*

[IBSSC '21] Ashwini Dalvi, Irfan Siddavatam, **Viraj Thakkar**, Apoorva Jain, Faruk Kazi, Sunil Bhirud. "Link Harvesting on the Dark Web." *IEEE Bombay Section Signature Conference (IBSSC)*, pp. 1-5. IEEE, 2021. [URL](#)

ACADEMIC POSTERS AND WORK-IN-PROGRESS

[FAST '24] Madhumitha Sukumar, Jiaxin Dai, Kaushiki Singh, Vikriti Lokegaonkar, **Viraj Thakkar**, Zhichao Cao. "LLM-assisted Automatic-Configuration and Tuning Framework for LSM-based Key-Value Stores." *22th USENIX Conference on File and Storage Technologies, 2024.*

TECHNICAL SKILLS

Languages: Python, C/C++, Bash (Shell), SQL, MATLAB, Front End Development (HTML, CSS, JS)

Tools & More: Git, Kubernetes, Docker, Django, Pandas, TensorFlow, AWS, Azure, GCP, IBM Quantum

Databases & OS: PostgreSQL, RocksDB, Neo4j, MySQL, SQLite, Linux/Unix

Soft Skills: Project Management, Leadership, Consistency, Communication Skills, Creative problem solving.

AWARDS AND ACHIEVEMENTS

FAST '24 Travel Grant 2024

Stack Overflow 2023

- 1000+ Reputation with 50+ answers. **Top 15%** contributor in 2023.

Hackathon by SVB & ASU, Tempe, AZ 04/2022

- Developed full stack web application to connect Investors with Startups. Achieved second place in the event.

Qubit by Qubit, a course by IBM Quantum and The Coding School 2021

Internship and Hackathon Organizer, Mumbai, IN 08/2020 - 04/2021

- Organized and coordinated a college-wide internship activity alongside college faculties. Created and managed task assignments and updated for 100+ interns over 2 semesters.
- Worked alongside college's computer science society for organizing and managing College Hackathon.

Team KJSCE Robocon, Mumbai, IN 08/2018 - 08/2020

- Spearheaded a team of 40+ individuals and mentored members to take over the team in coming years.
- Achieved National Rank 8 and Winner of Best Solution Idea amongst 50+ teams in "DD-Robocon 2020".